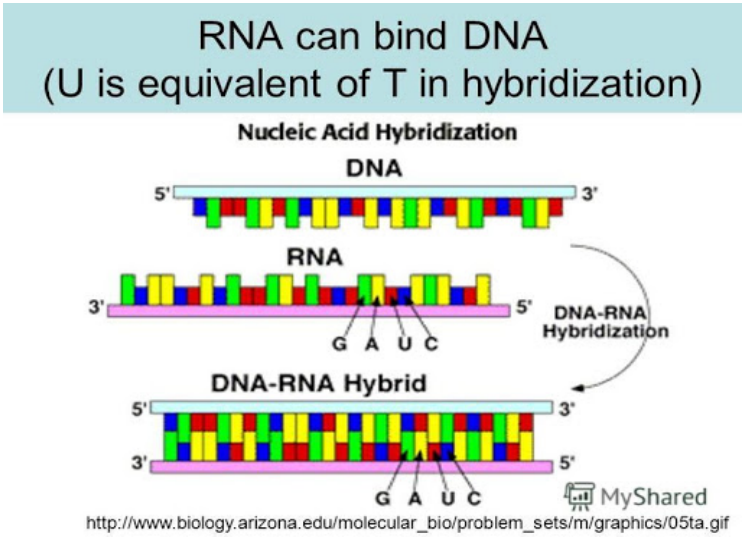
Chapter 4
Blotting methods

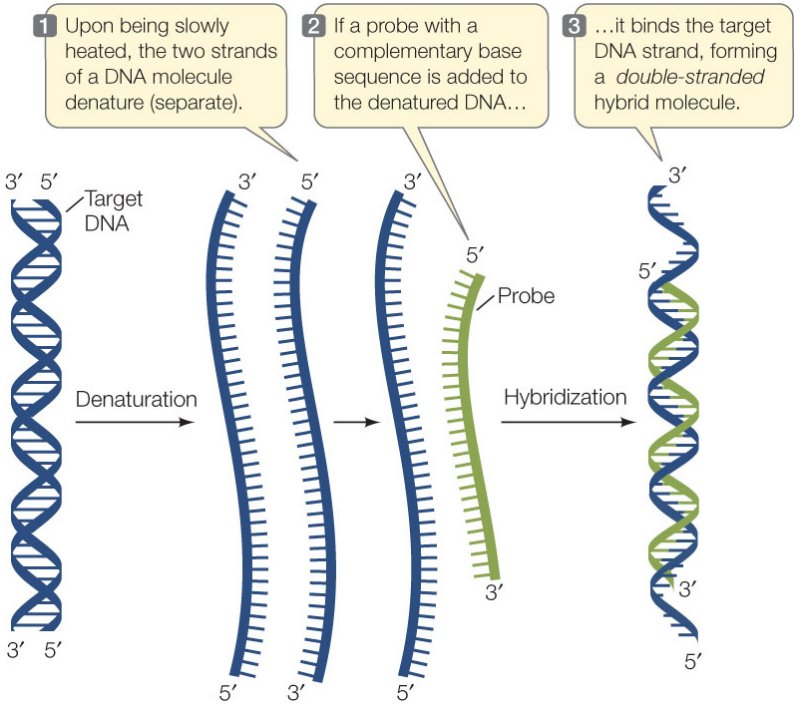
I. Introduction
II. Southern blotting
III. Northern blotting
IV. In situ hybridization (ISH)

I. Introduction

- Blotting methods are molecular biology techniques used to detect specific biomolecules—such as DNA, RNA, or proteins—after they have been separated by gel electrophoresis.

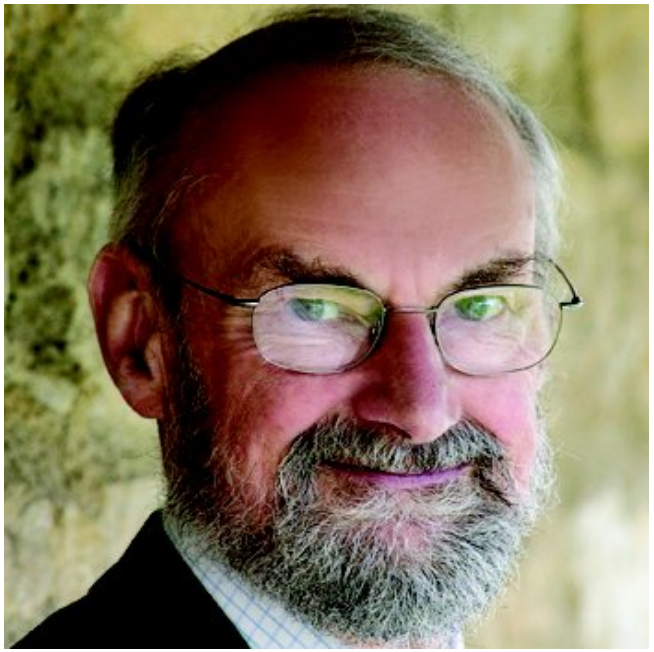
- DNA – DNA, RNA – DNA hybridization
- Probe



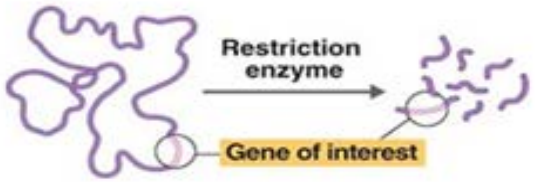


II. Southern blotting

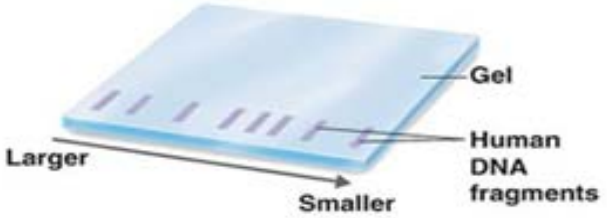
- Purpose: Detects specific DNA fragments
- 1975: Edwin Southern
- Principles:
  - Quantified DNA
  - Denatured DNA
  - Hybridized DNA-DNA



Principle and process



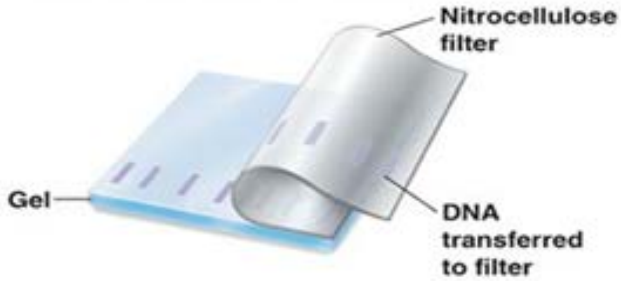
1 DNA containing the gene of interest is extracted from human cells and cut into fragments by restriction enzymes.



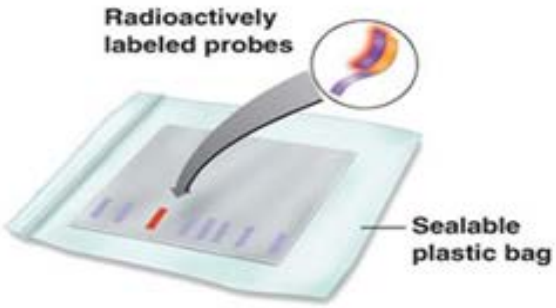
2 The fragments are separated according to size by gel electrophoresis. Each band consists of many copies of a particular DNA fragment. The bands are invisible but can be made visible by staining.



3 The DNA bands are transferred to a nitrocellulose filter by blotting. The solution passes through the gel and filter to the paper towels.



4 This produces a nitrocellulose filter with DNA fragments positioned exactly as on the gel.



5 The filter is exposed to a radioactively labeled probe for a specific gene. The probe will base-pair (hybridize) with a short sequence present on the gene.



6 The filter is then exposed to X-ray film. The fragment containing the gene of interest is identified by a band on the developed film.

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Applications

- Gene mapping and identification.
- Detection of insertions, or deletions.

III. Northern blotting

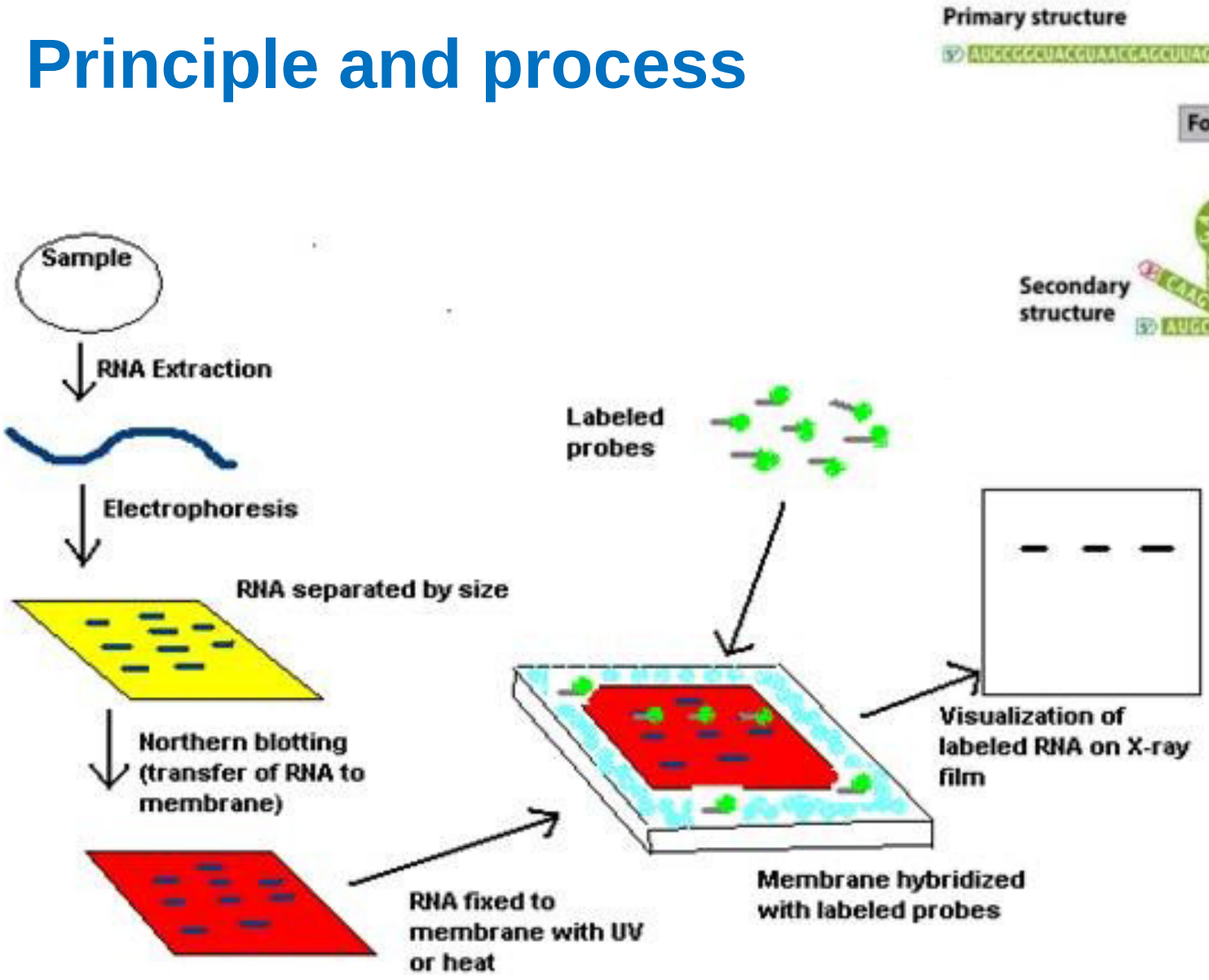
- **Purpose:** Detects and measures RNA (especially mRNA) expression levels.
- **1977:** James Alwine, David Kemp, and George Stark at Stanford University





- Principles:
  - Quantified RNA
  - Denatured RNA
  - Hybridized RNA-DNA

Principle and process



Applications

- Analysis of gene expression patterns.
- Detection of RNA virus

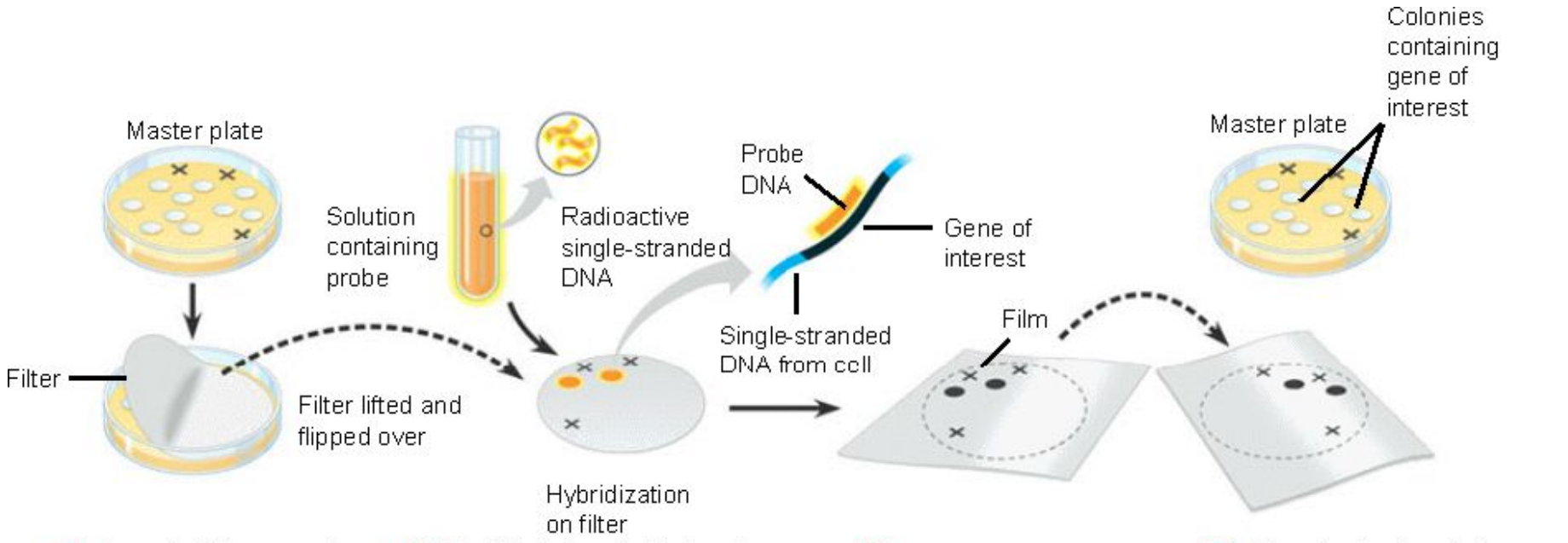
Compare

- **Southern blot v.s. Northern blot**
  + RE
  + Denatured RNA
  + Denatured dsDNA
  + DNA-DNA, RNA-DNA

- **Southern, Northern blot v.s. PCR**
  - conventional PCR; real time PCR

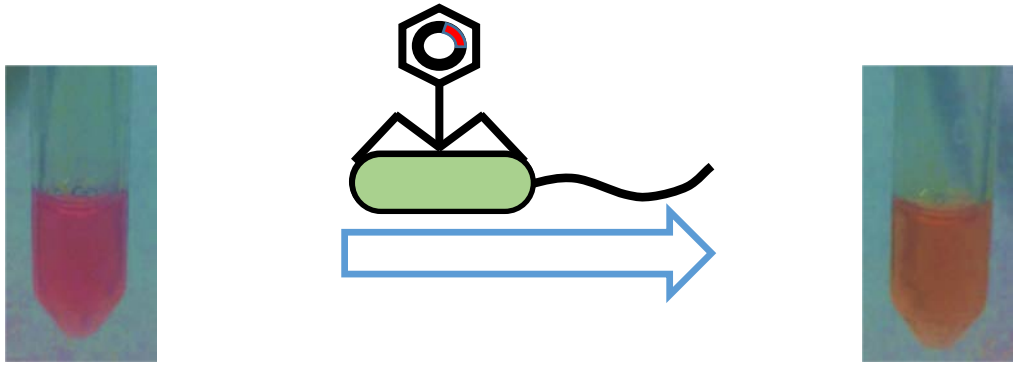
IV. In situ hybridization (ISH)

- ISH is base pairing (hybridization) between a probe and its complementary sequence within fixed tissue or cell samples.



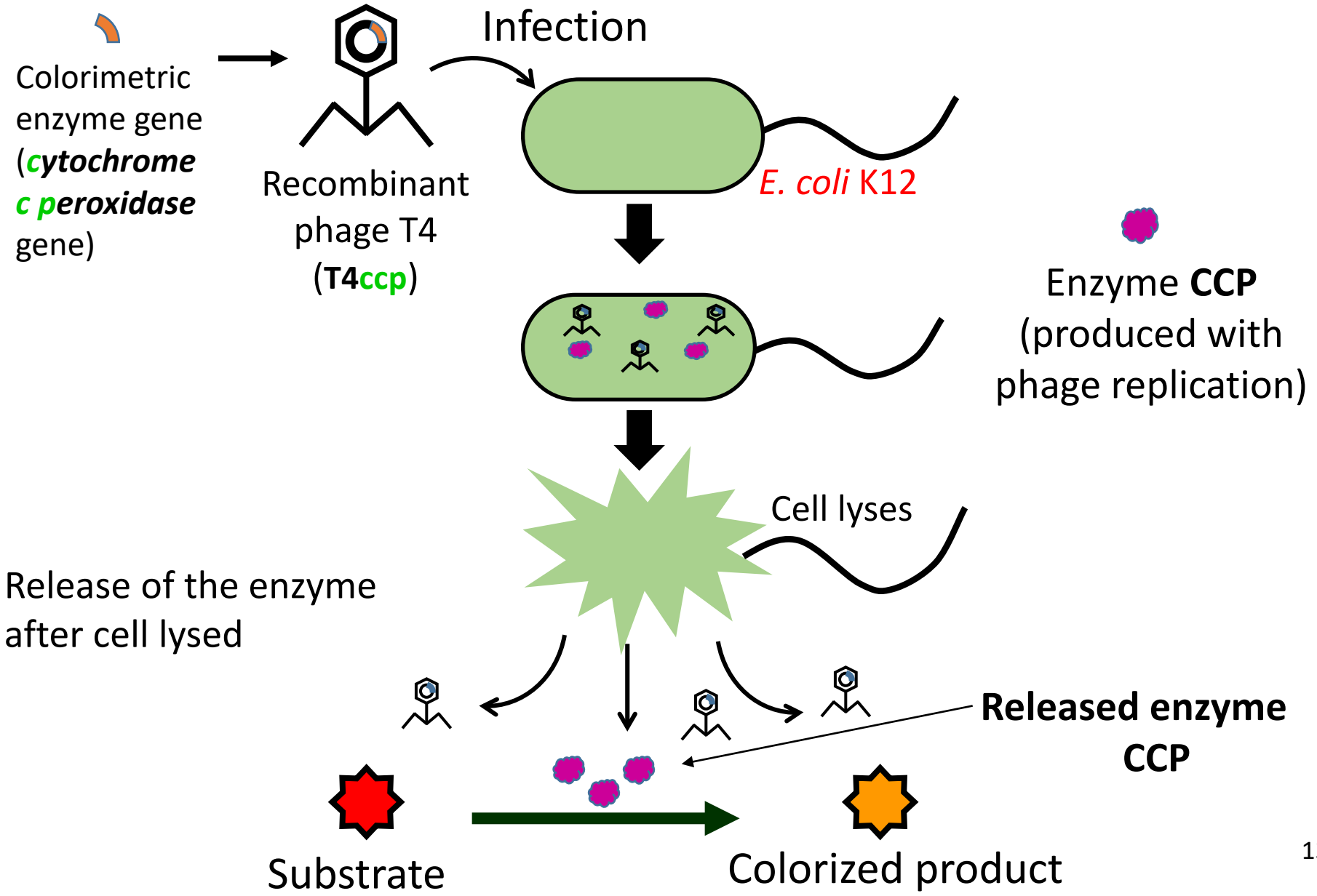
Example

- To develop a phage-based colorimetric method for the detection of *E. coli* K12

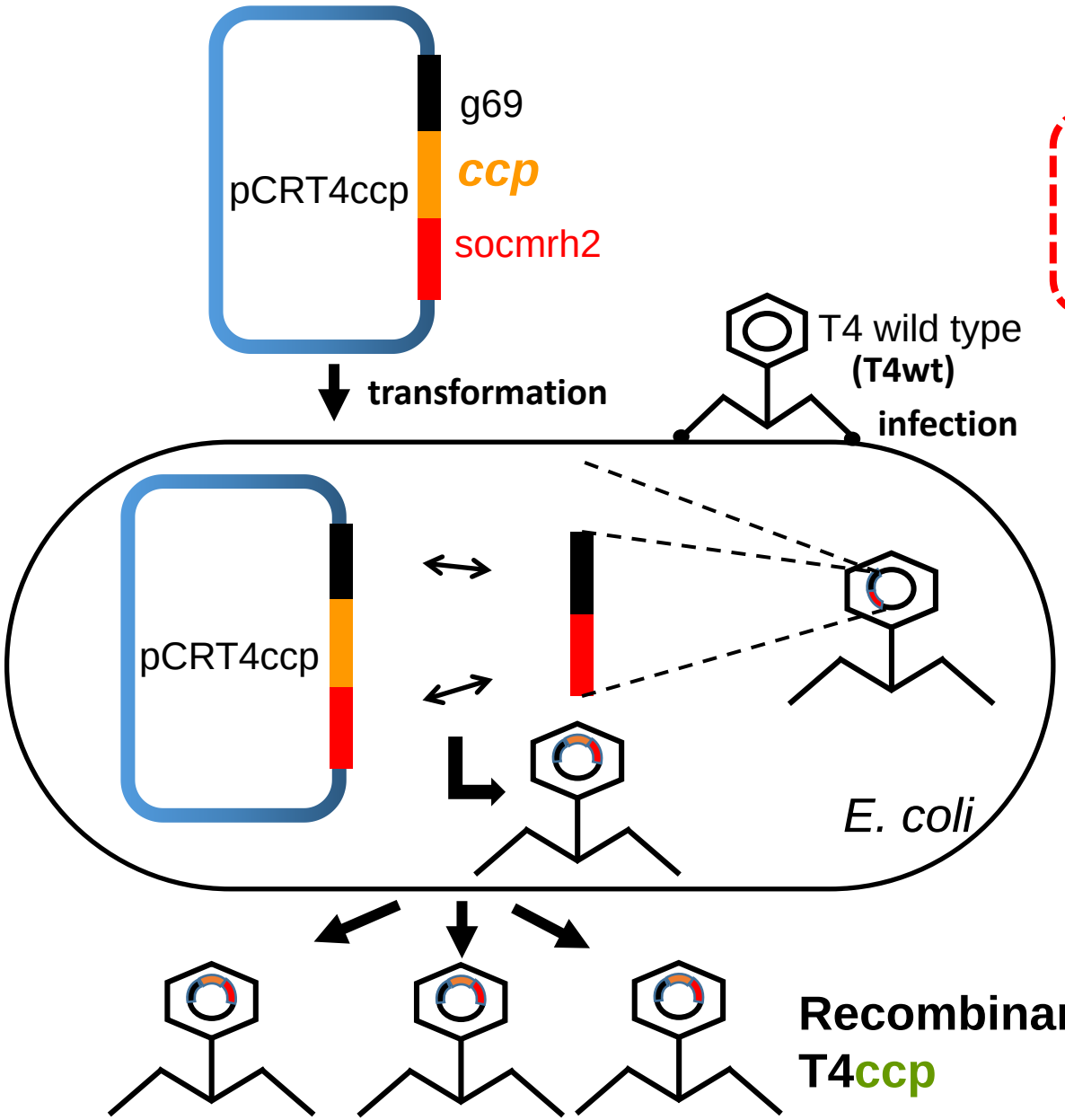


- In construction of the new method: **T4** phage (and its host ***E. coli* K12**) were used since T4 phage has been best studied.

Principle of the detection



Construction of recombinant phage

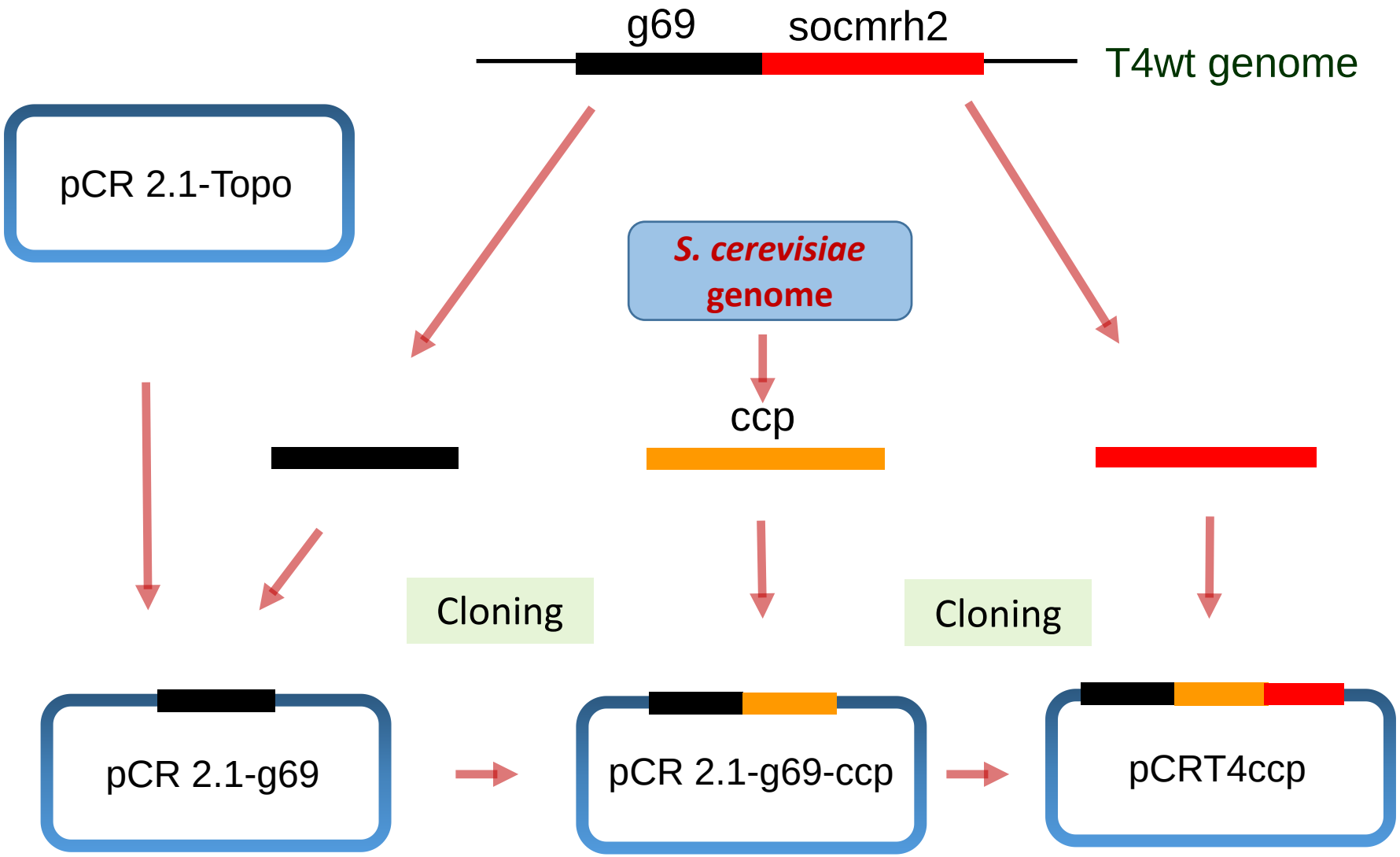


Construction of a plasmid vector

Recombination

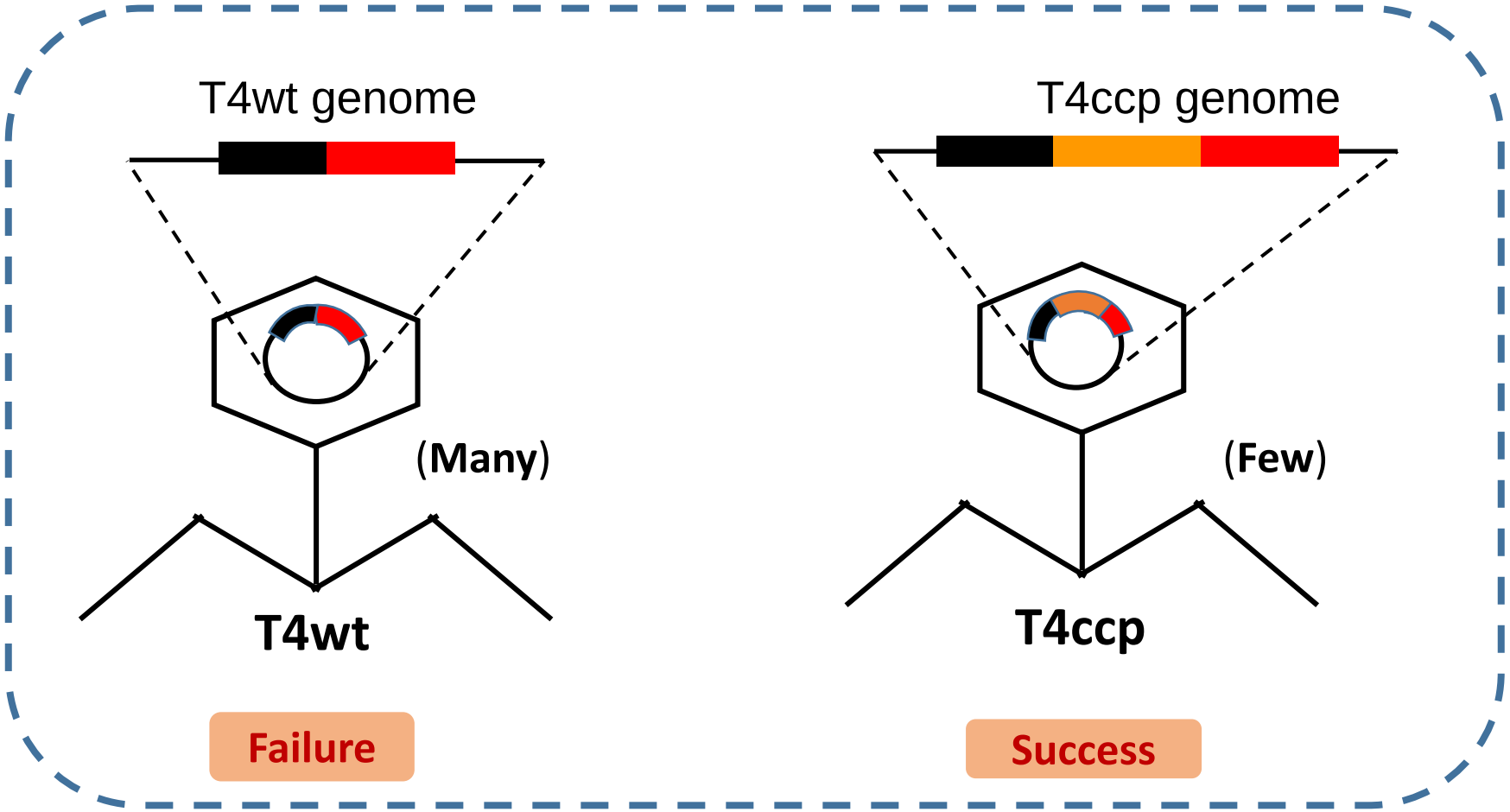
Isolation

Construction of the vector



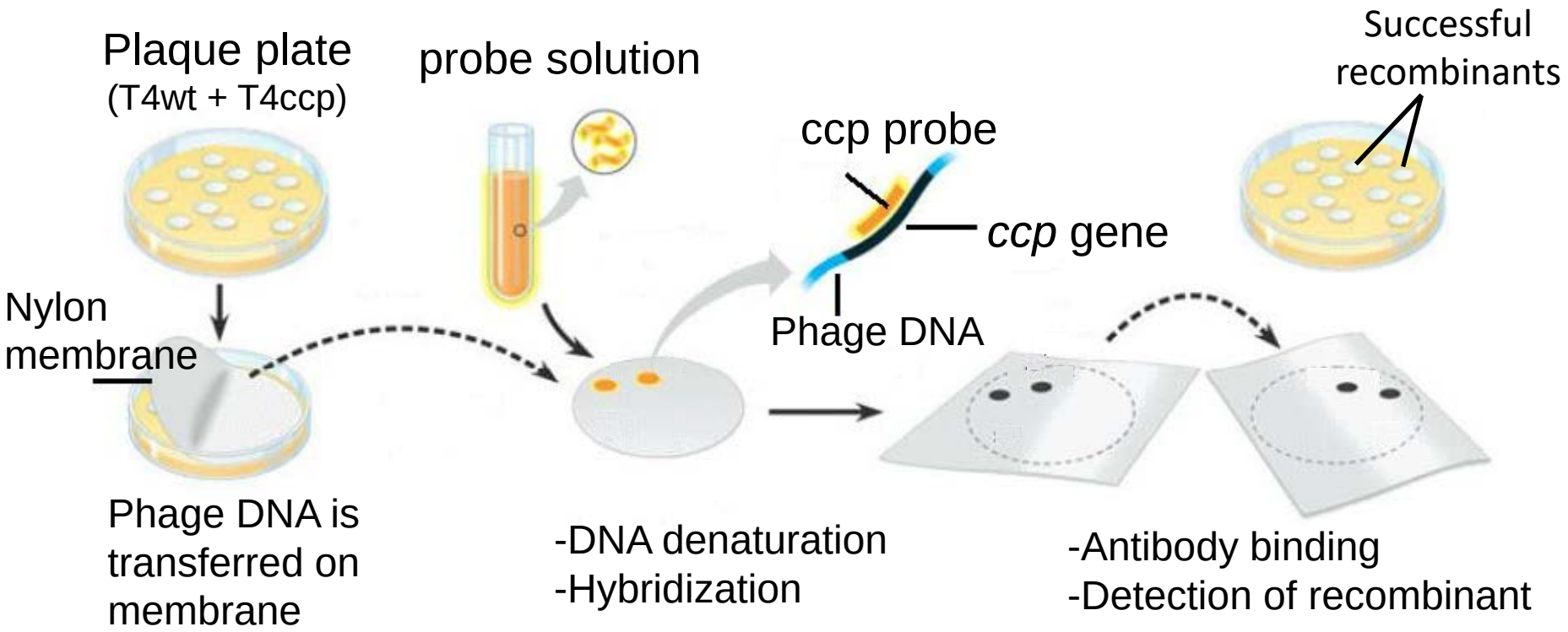
Homologous recombination

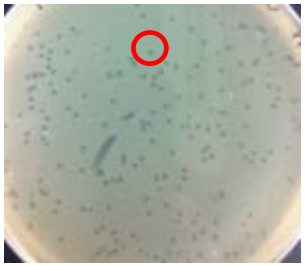
Produced phage



Isolation of recombinant phage

Hybridization technique: used to isolate the recombinant phage





Plaque plate

Hybridization



Membrane